

ECEN 215, Spring 2024

Electrical Circuit Theory

Lecture 5: Resistive Circuits Revision

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Assignments

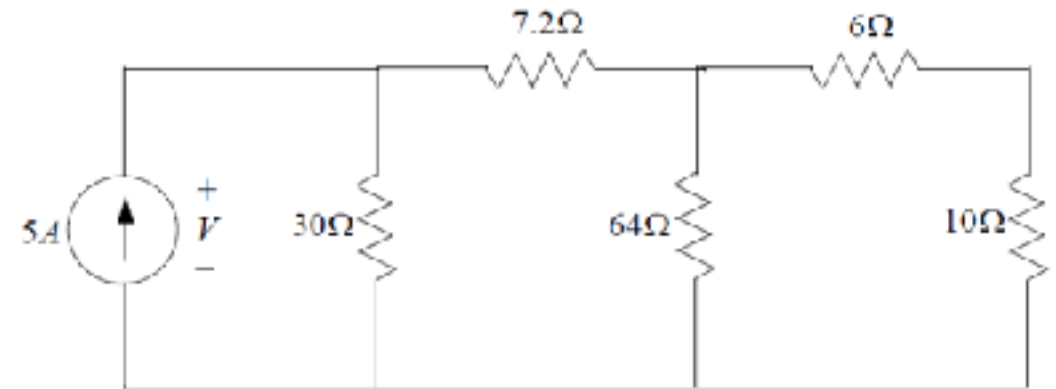


- Exam #1 covering chapter 1, 2 will be this Wednesday (Feb 21) in classroom, 9:10 to 10:00
- Please study all notes and assignments
- I will provide a cheat sheet

Example



- For the circuit shown:
 - (a) Find the voltage V ,
 - (b) Find the power delivered to the circuit by the current source,
 - (c) Find the power dissipated in the 10Ω resistor.



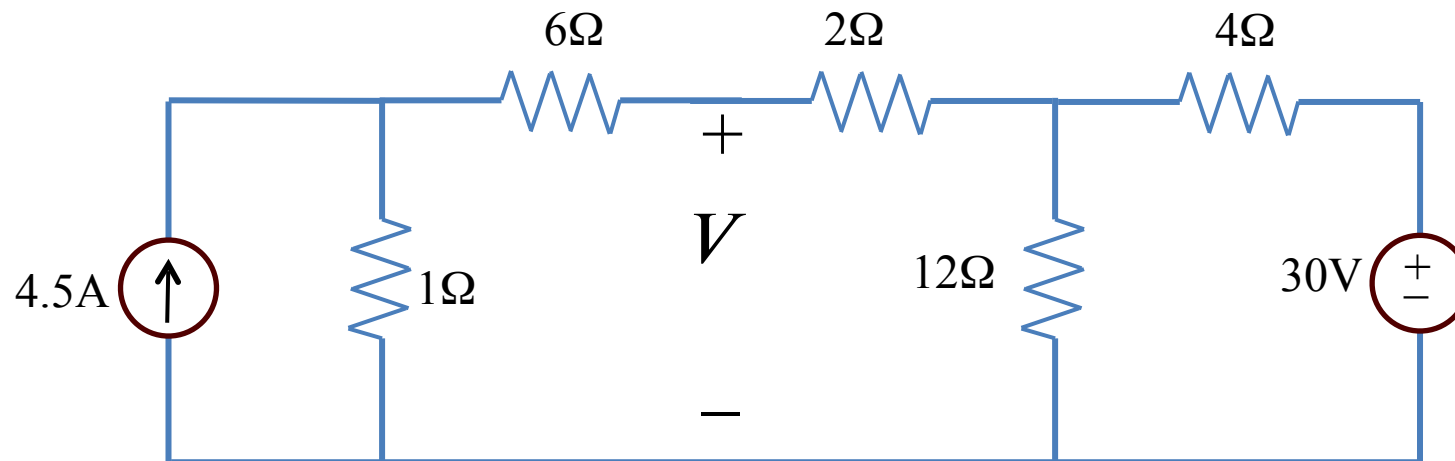
Example



Example 2



- Use the node-voltage method to find V in the circuit.



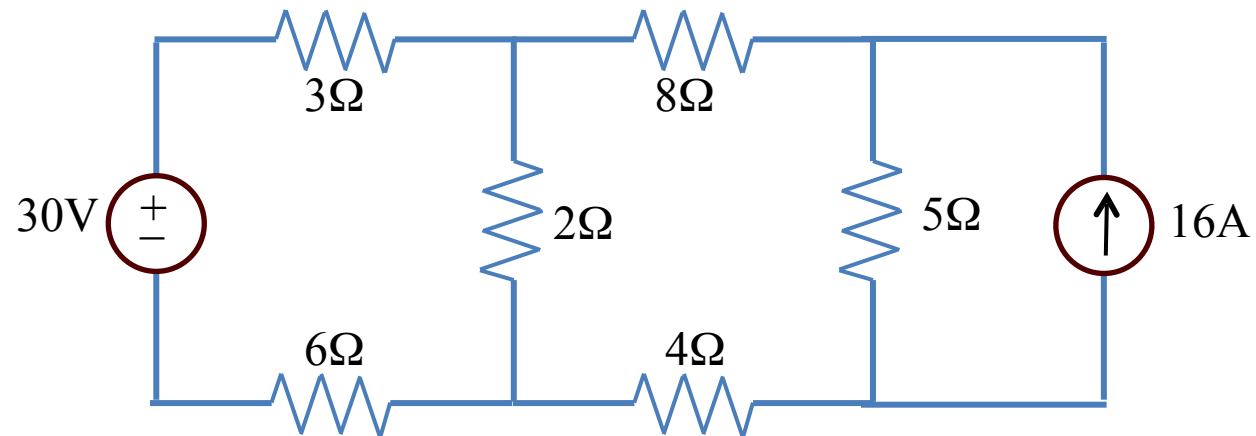
Example 2



Example 3



- Use the mesh-current method to find the power dissipated in the 2Ω resistor.



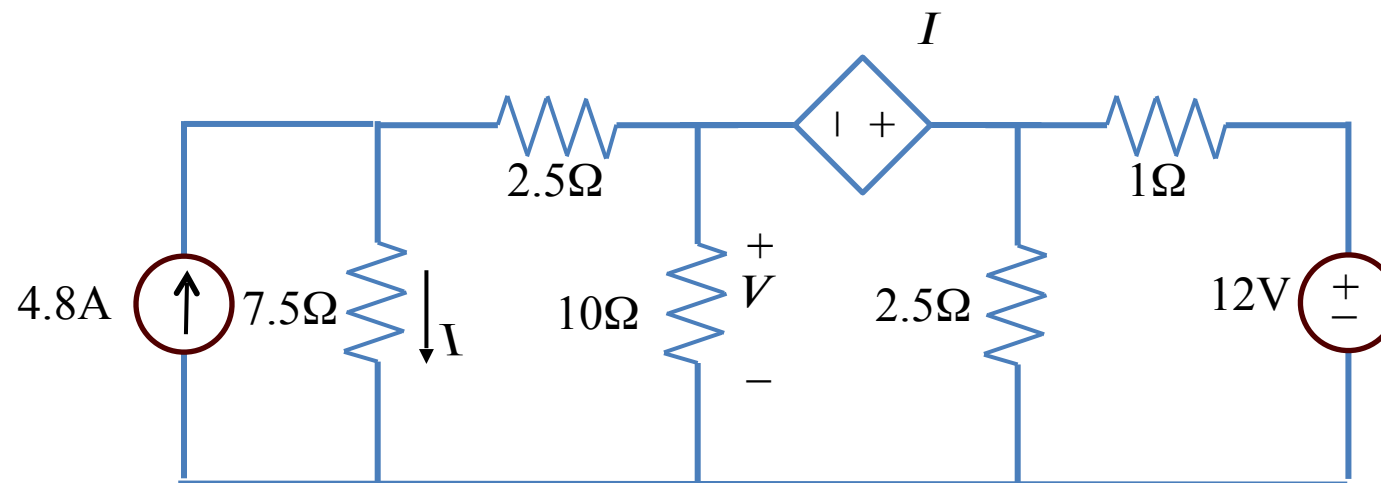
Example 3



Example 4



- Solve for voltage V

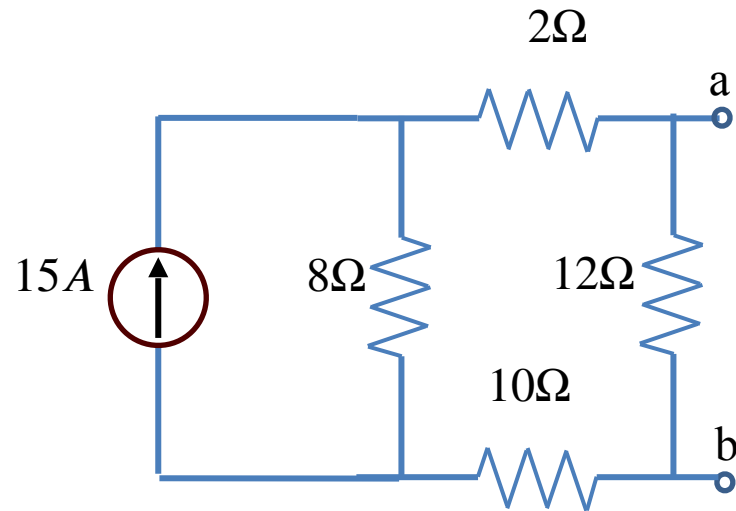




Example 5



- Find the Thevenin and Norton equivalent circuits with respect to the terminals a-b.



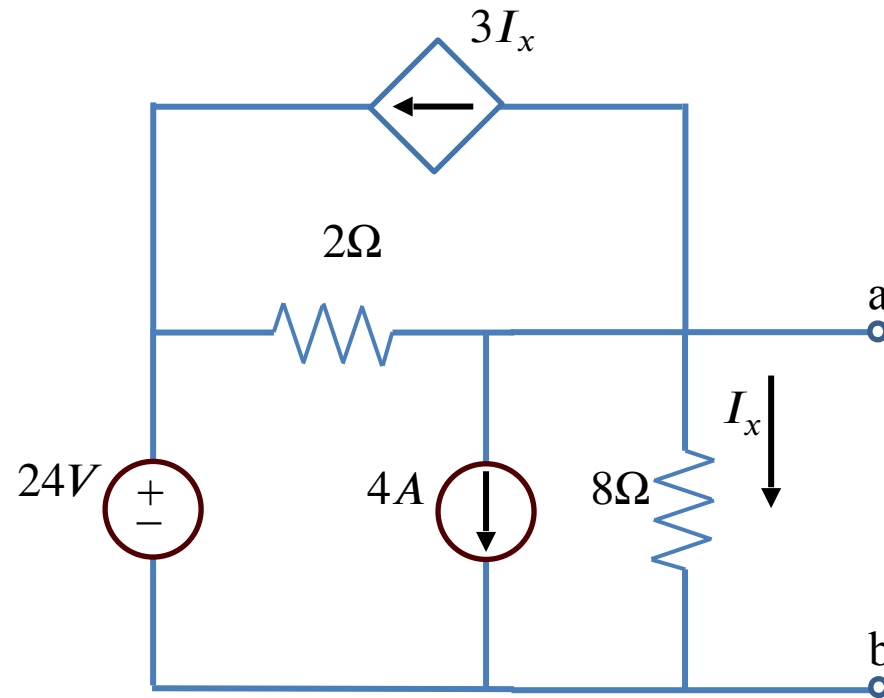
Example 5—solution



Example 6



- Find the Thevenin equivalent with respect to the terminals a-b.



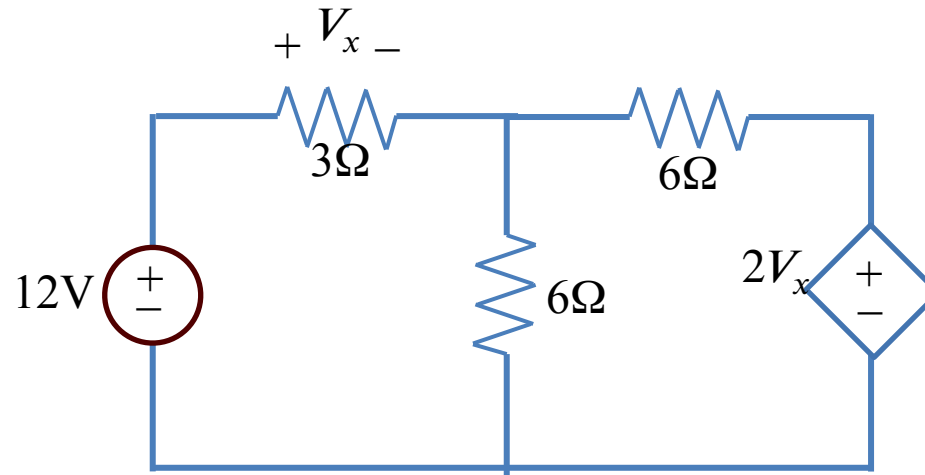
Example 6—solution



Example 7



- Find V_x by applying a sequence of source transformations.

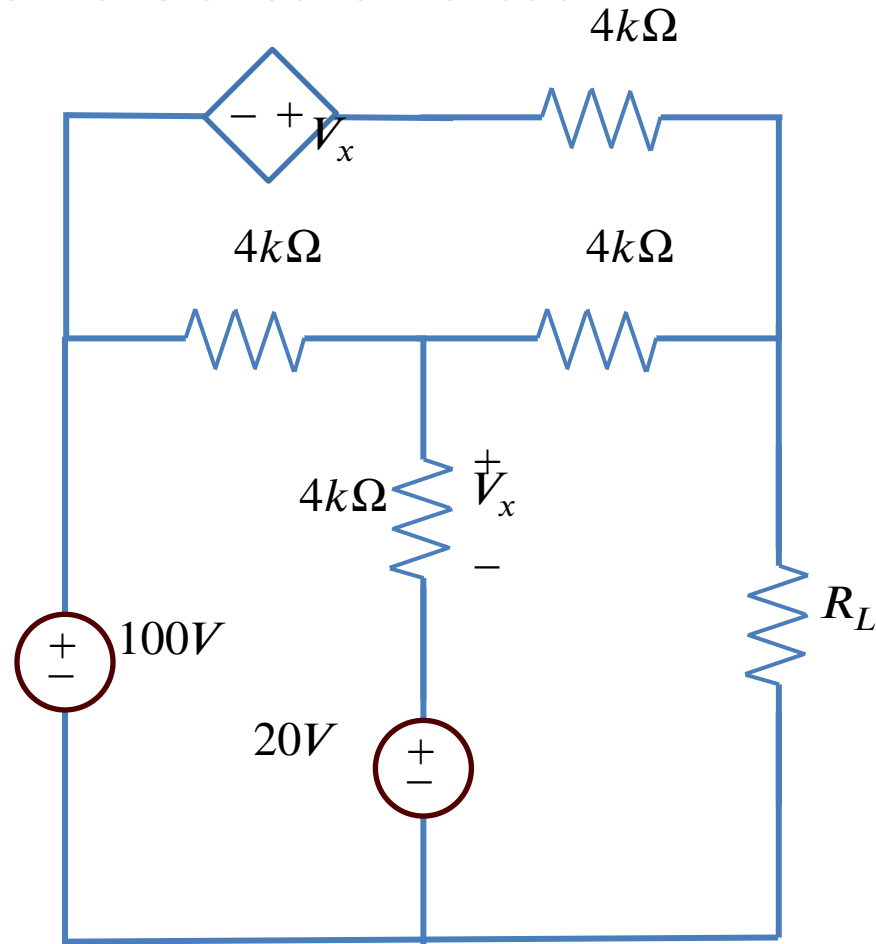


- Note:** If the circuit has dependent sources, we need to make sure to preserve the element with the control voltage/current.

Example 8



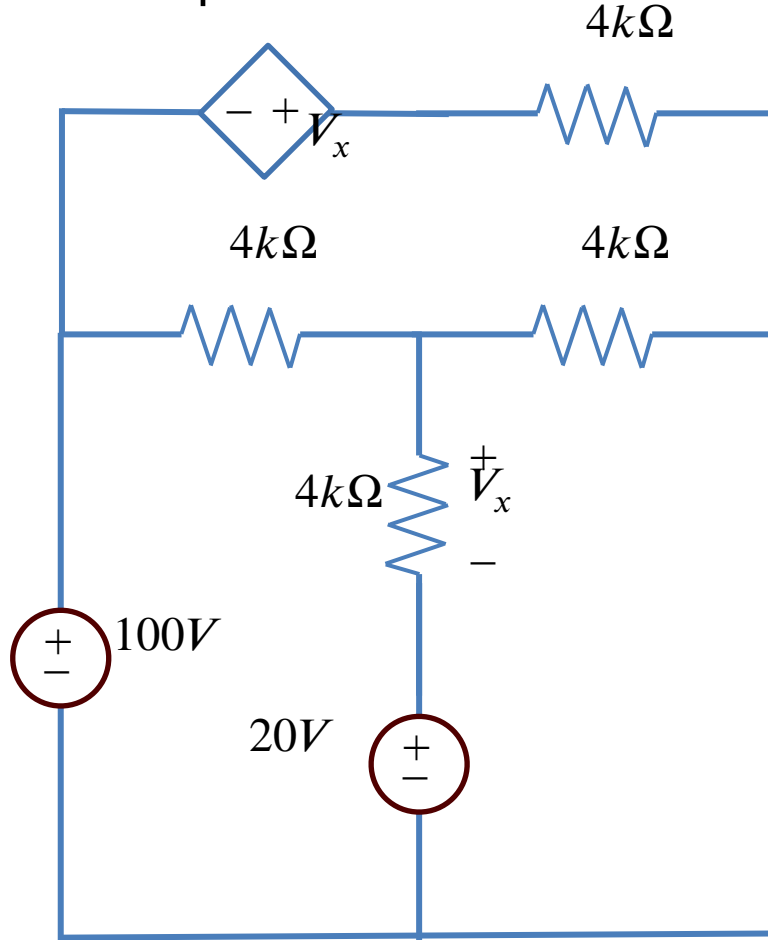
- (a) Find R_L that maximizes power transfer to the load.
(b) What is the maximum power transferred to the load?



Example 8, cont.



- (a) Find R_L that maximizes power transfer to the load.
(b) What is the maximum power transferred to the load?

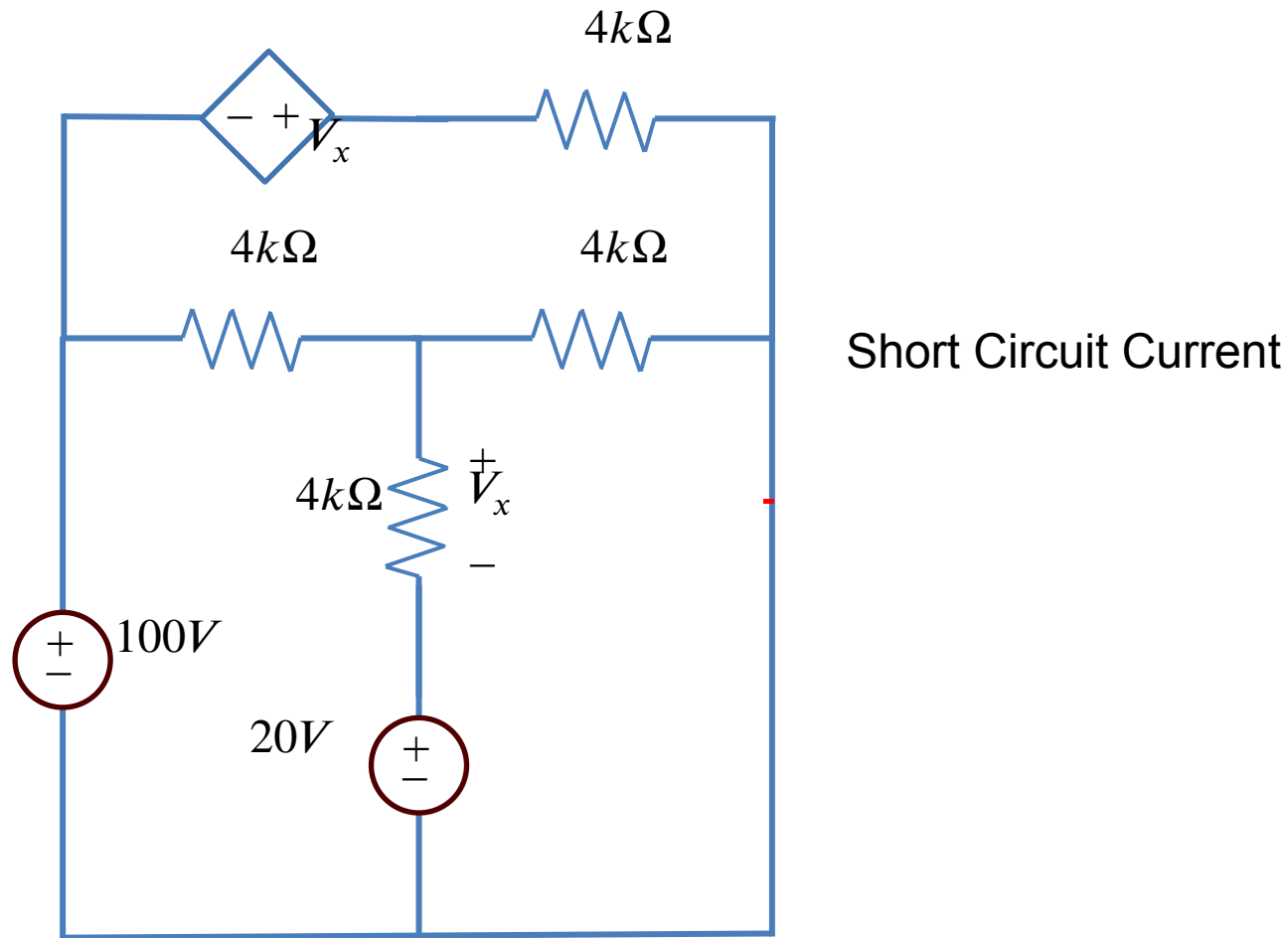


Open Circuit Voltage

Example 8, cont.



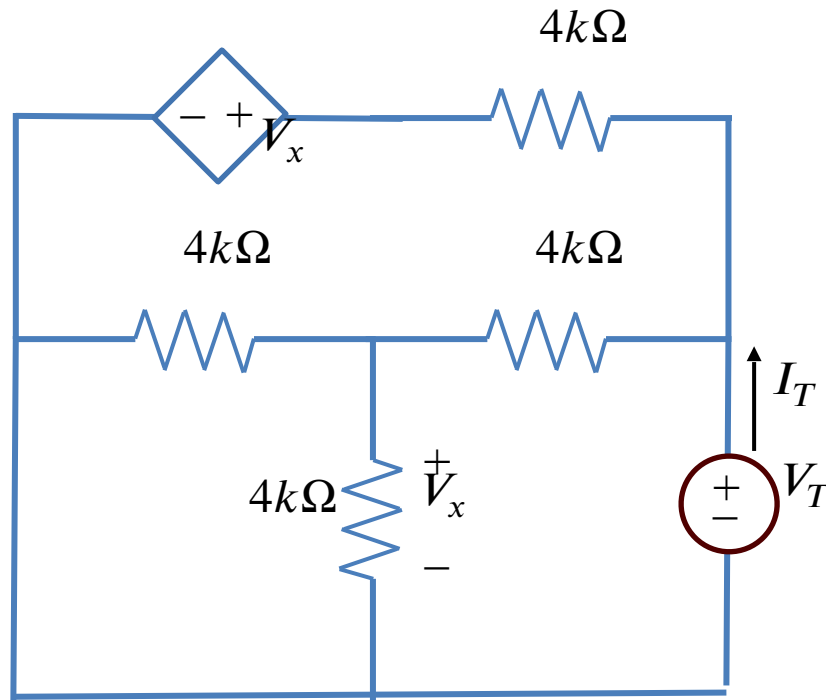
- (a) Find R_L that maximizes power transfer to the load.
 (b) What is the maximum power transferred to the load?



Example 8, cont.



- (a) Find R_L that maximizes power transfer to the load.
 (b) What is the maximum power transferred to the load?



Alternatively (Method 3)
 Disable Independent Sources
 and Apply Test Source